STAT 131: Lecture Notes 2

Winter 2024

2.1 Counting methods [DS 1.7, 1.8]

Motivated by equally likely outcome spaces, we will now discuss commonly used methods of counting in this subsection. These are tools that can be used to compute probabilities of events. Most classical probability calculations can be done using these methods.

Theorem 2.1 (The multiplication rule)**.** *Consider an experiment with multiple stages (k total stages); in stage i, there are n_i options. Then, in total, the number of options for the experiment is*

$$\prod_{i=1}^{K} n_i$$

Example 2.1. What is the total number of outcomes?

- rolling two dice $K=2$, $n_1=6$, $n_2=6$, $\prod_{i=1}^{K} n_i = 6\times 6 = 36$
- toss one coin *and* roll one die $K=2$, $n_1=2$, $n_2=6$, $\prod_{i=1}^{K} n_i = 2\times 6 = 12$
- Phone numbers $K=7$, $n_i=10$ $\forall i$, $\prod_{i=1}^{K} n_i = 10^7$

♢

2.1.1 Sampling with and without replacement

Definition 2.1 (Sampling with replacement)**.** Select k items from a total of n items with replacement.

Example 2.2. Suppose a box contains n numbered balls; for each selection, record the number and put the selected ball back to the box. If you repeat this two times, you can get the same number twice. ♢

Example 2.3 (roll two fair dice)**.** The experiment that two fair dice are rolled can be thought of as the same die being independently rolled twice. ♢

Definition 2.2 (Sampling without replacement)**.** Select k items from a total of n items without replacement.

Example 2.4 (Texas Holdem)**.** 3 common cards are dealt sequentially from a deck of cards. 2 cards are dealt to each player from the same deck. ♢

Example 2.5 (Blackjack)**.** 2 cards are dealt to each player from the same pile of cards. ♢

Example 2.6 (Lotto numbers)**.** Lotto numbers are drawn sequentially. ♢

2.1.2 Permutation, Combinations, and Partitions

We will learn three tools for counting. This is called **Combinatorics**, and is a branch of mathematics that we utilize.

1. Permutations
2. Combinations
3. Partitions

First, we define a core counting operation called the factorial.

Definition 2.3 (Factorial)**.** Factorial: $n!$ is defined as

$$n! = n \times (n-1) \times ... \times 2 \times 1.$$

Also, by definition, $0! = 1$.

A factorial is useful for counting permutations.

Definition 2.4 (All permutations)**.** The number of distinct orderings of n elements is $n!$.

Now, suppose you are at Baskin Robins building an ice cream cone. The shop has n flavors of ice cream, but you only want $k < n$ scoops. How many different cones are possible if each flavor can only be used once, and the order of the flavors matters?

There are n choices for the bottom flavor, $(n-1)$ for the next flavor and so on... with $n-(k-1)$ for the final k^{th} flavor

$$n \cdot (n-1) \cdot (n-2) \cdots (n-(k-1)) = \frac{n(n-1)(n-2)\cdots 2\cdot 1}{(n-k)!} = \frac{n!}{(n-k)!}$$

Each ordering here is called a permutation.

Next, we define the permutation number $P_{n,k}$. This represents the number of ways to order k elements chosen from n elements; also called k-permutation.

Definition 2.5 (Permutation number)**.** The Permutation number $P_{n,k}$ is the number of distinct orderings such that k elements are selected from n items **in order**, without replacement.

$$P_{n,k} = n!/(n-k)!$$

Note: Order matters!

Note, the number of ways to choose n out of n elements is $P_{n,n}$ simplifies to n factorial $(n!)$. In this case, no choosing occurs; we only need to count the ways to order.

Example 2.7 (Choosing officers)**.** Suppose a club has 25 members, a president and a secretary are to be chosen from the membership. How many distinct configurations can we have?

25 options for president, 24 options for secretary (pres. cannot be secretary)

$n = 25$, $k = 2$

$$P_{25,2} = \frac{25!}{(25-2)!} = \frac{25 \cdot 24 \cdot 23 \cdots 1}{23 \cdot 22 \cdots 1} = 25 \cdot 24$$

♢

Example 2.8 (Bookshelf)**.** Suppose you have 10 books on your bookshelf, how many ways are there to arrange all of the books?

$n = 10$ $\quad$ $k = 10$

$$P_{10,10} = \boxed{10!}$$

♢

Example 2.9 (Birthday problem)**.** What is the probability that at least two of the students in STAT131 have the same birthday? We can assume that there are 365 days in a year, and being born on any given day is equally likely.
Hint: $E = \{$at least two of the students have the same birthday$\}$ and we are interested in $\mathbb{P}(E)$.

$E^c = \{$ no students share a birthday$\}$ and $P(E) = 1 - P(E^c)$

suppose 100 students

$$P(E^c) = \frac{|E^c|}{|\Omega|} = \frac{P_{365,100}}{|\Omega|} \qquad \text{but } |\Omega| = 365^{100}$$

$$= \frac{P_{365,100}}{365^{100}} \Rightarrow P(E^c) = 3 \times 10^{-7} \Rightarrow P(E) = 1 - P(E^c) = 0.9999997$$

♢

Approximate factorials Most computers cannot handle $n!$ and arithmetic operations when n is big. The trick is to complete the arithmetic operations in log-scale and exponentiate the results in the end. To compute $\frac{P_{n,k}}{n^k}$, we compute $\log(P_{n,k}) - k\log(n)$ where $\log(P_{n,k}) = \log(n!) - \log((n-k)!) = \sum_{i=n-k+1}^{n} \log(i)$. Identity: $\frac{P_{n,k}}{n^k} = e^{\log(P_{n,k}) - k\log(n)}$ where $\log(\cdot)$ is the natural log.

$$P(E^c) = \frac{P_{365,100}}{365^{100}} = e^{\log(365!) - \log(265!) - 100\log(365)}$$

Consider the Baskin Robins ice cream problem again. How many scoops are possible if the order of the flavors in the cone does not matter?

The number of permutations over counts! $\left(\binom{c}{v} = \binom{v}{c}\right)$

If we chose K flavors, there are K! ways to order them and they are all equivalent. Thus, we must divide the number of permutations by K!

$$\left(\frac{n!}{(n-K)!}\right) \Big/ K! = \frac{n!}{(n-K)!\,K!}$$

In this case, each choice of k flavors is called a combination.

Definition 2.6 (Combination number)**.** Suppose a set has n elements and we select k elements without replacement. Each **subset** of size k is a *combination* of n elements taken k at a time. The number of distinct combinations:

$$C_{n,k} = \frac{P_{n,k}}{k!} = \frac{n!}{(n-k)!k!}.$$

Also known as binomial coefficients $\binom{n}{k}$. **Note: Order doesn't matter!**

Example 2.10. Evaluate $S_1 = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}$.

Imagine you have n elements. S_1 is the total number of possible subsets.

All subsets of size 0 + all subsets of size 1 + ...

But the total number of possible subsets is 2^n

$$S_1 = 2^n$$

♢

Example 2.11. Evaluate $S_2 = \binom{n}{0}y^n + \binom{n}{1}xy^{n-1} + \binom{n}{2}x^2y^{n-2} + \cdots + \binom{n}{n}x^n$

Consider the expansion $(x+y)^n$

The term $x^k y^{n-k}$ appears exactly $\binom{n}{k}$ times.

So $S_2 = (x+y)^n$

♢

Theorem 2.2 (Binomial theorem)**.**

$$(x+y)^n = \sum_{i=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Example 2.12. Complete square: $(x+y)^2 =?$ and $(x+y)^3 =?$.

$$(x+y)^2 = \binom{2}{0}y^2 + \binom{2}{1}xy + \binom{2}{2}x^2 = y^2 + 2xy + x^2$$

$$(x+y)^3 = \binom{3}{0}y^3 + \binom{3}{1}xy^2 + \binom{3}{2}x^2y + \binom{3}{3}x^3$$

$$= y^3 + 3xy^2 + 3x^2y + x^3$$

♢

Example 2.13. Evaluate $S_3 = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n$.

This is the same as S_2 with $y=1$

So $S_3 = (x+1)^n$

♢

Lemma 2.3. *A few frequently used properties of binomial coefficients:*

- $\binom{n}{0} = \binom{n}{n} = 1$,
- $\binom{n}{k} = \binom{n}{n-k}$.
- $\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$.
- *Vandermonde's identity:*

$$\binom{m+n}{k} = \sum_{j=0}^{k} \binom{m}{j}\binom{n}{k-j}.$$

To understand Vandermonde's identity, suppose you are trying to pick k leaders from a group of m women and n men. We know that there are $m+n$ members of the group, thus there are $\binom{m+n}{k}$ ways to pick the k leaders. This is the left-side of Vandermonde's identity. However we can also view the problem in a different way. We first pick j leaders from from the set of m women ($0 \le j \le k$) and then we pick $k-j$ leaders from the set of n men. This can be done for any of the possible values of j, so we must sum over the possible values of j. This is the right-side of Vandermonde's identity.

Example 2.14 (Poker)**.** A 5-card hand is dealt from a standard, well-shuffled 52-card deck. The hand is called a *full house* in poker if it consists of three cards of some rank and two cards of another rank. e.g. three K's + two 7's, where ordering does not matter. What is the probability of a full house?

$P(\text{full house}) = \dfrac{|\text{full house}|}{|\Omega|}$ $\quad |\Omega| = \binom{52}{5}$

$|\text{full house}| = 13 \cdot \binom{4}{3} \cdot 12 \cdot \binom{4}{2}$ ← ways to choose 2 out of 4; ← second rank; ← ways to choose 3 out of 4; ← first rank

So $P(\text{full house}) = \dfrac{13\binom{4}{3}\cdot 12\cdot\binom{4}{2}}{\binom{52}{5}}$

$\simeq 0.001441$

♢

Example 2.15 (Fair coin tossed)**.** A fair coin is to be tossed 10 times. Determine the probability that exactly 3 heads will occur? Also determine the probability that three or fewer heads will occur?

$P(3 \text{ heads}) = \dfrac{|3 \text{ heads}|}{|\Omega|} = \dfrac{\binom{10}{3}}{2^{10}}$

$P(3 \text{ or fewer heads}) = P(0 \text{ heads}) + P(1 \text{ head}) + P(2 \text{ heads}) + P(3 \text{ heads})$

$= \frac{\binom{10}{0}}{2^{10}} + \frac{\binom{10}{1}}{2^{10}} + \frac{\binom{10}{2}}{2^{10}} + \frac{\binom{10}{3}}{2^{10}} = \frac{\binom{10}{0}+\binom{10}{1}+\binom{10}{2}+\binom{10}{3}}{2^{10}}$

♢

Example 2.16 (Sampling without replacement)**.** Suppose an urn contains 15 yellow and 30 blue balls, from which 10 balls will be selected at random. What is the probability that exactly three yellow balls will be selected?

$P(\text{3 yellow}) = P(\text{3 yellow and 7 blue})$

$= \frac{|\text{3 yellow and 7 blue}|}{|\Omega|}$

$= \frac{\binom{15}{3}\cdot\binom{30}{7}}{\binom{45}{10}}$

♢

Next, suppose we want to partition a set of n elements into k subsets so that the ith subset contains n_i elements, and $\sum_i n_i = n$. How many ways can this be done?

Definition 2.7 (Partitions)**.** The number of ways to partition n elements into r groups of size $n_1, \cdots, n_r$ is

$$\binom{n}{n_1, \cdots, n_k} = \binom{n}{n_1}\binom{n-n_1}{n_2}\binom{n-n_1-n_2}{n_3}\cdots\binom{n-n_1-\cdots-n_{k-2}}{n_{k-1}} = \frac{n!}{n_1!\,n_2!\cdots n_k!}$$

This is the *multinomial* coefficient.

Can you see how the combinations are a specific case of partitions, so that the binomial coefficient is a specific case of the multinomial coefficient?

Example 2.17. How many different words (even meaningless) can be obtained by rearranging the word PEPPER?

$n = 6 \quad n_1 = 3 \quad n_2 = 2 \quad n_3 = 1$

(P) (E) (R)

$\frac{n!}{n_1!\,n_2!\,n_3!} = \frac{6!}{3!\,2!\,1!} = 60$

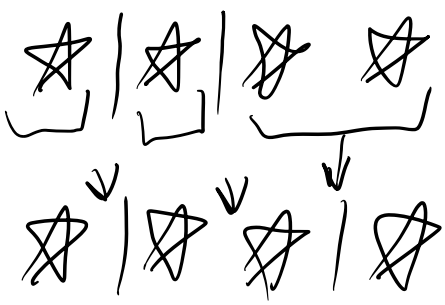
♢

Next, how many ways are there to put n identical balls into k numbered (distinguishable) bins? There is an elegant solution to this, called the stars and bars argument.

Example 2.18 (Stars and bars argument)**.** How to distribute n cookies to k kids?

- Each kid receives at least one cookie: $\binom{n-1}{k-1}$

$n=4$, $k=3$



$(1, 1, 2)$

$(1, 2, 1)$

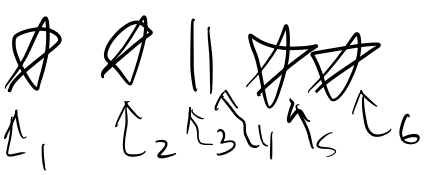
There are $n-1$ "gaps" where we can place $k-1$ bars

- Some kids may get zero cookies: $\binom{n+k-1}{k-1}$

$n=4$ $k=3$



$(1, 0, 3)$



$(2, 0, 2)$

There are $n+k-1$ "positions" of these we are selecting $k-1$ of the positions to be bars.

$\diamond$

2.2 Conditional Probability [DS 2.1]

Compare the following scenarios:

- COVID infection rate
- COVID infection rate if the person is fully vaccinated.

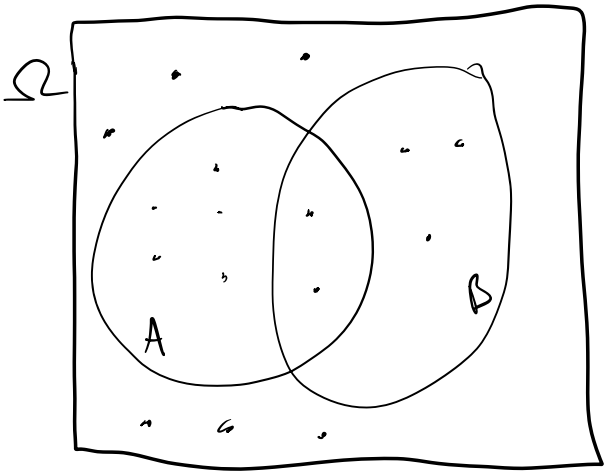
The two probabilities should be different. When more information is available, the probability calculation should reflect it.

Definition 2.8 (Conditional probability)**.** The probability of event A given event B has occurred is denoted as $\mathbb{P}(A|B)$. If $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

We can interpret $\mathbb{P}(A)$ as the "prior" probability of A without any other information. In contrast, we interpret $\mathbb{P}(A|B)$ as the posterior probability of A given B has occurred.

Example 2.19. Visualizing conditional probability:



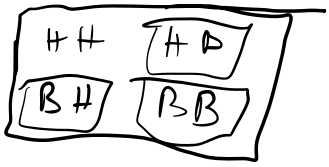
$$P(A) = \frac{7}{16} \qquad P(B) = \frac{5}{16}$$

$$P(A|\Omega) = \frac{P(A \cap \Omega)}{P(\Omega)} = \frac{P(A)}{1} = P(A)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{(2/16)}{(5/16)} = 2/5$$

♢

Example 2.20 (Two Offspring)**.** The offspring of a cow is called a calf. A calf is equally likely to be a heifer or a bull. Suppose that Cindy the Cow has two calves and we know that at least one is a bull. What is the probability that both calves are bulls?



$$P(\text{both bulls} \mid \text{at least 1 is a bull})$$

$$= \frac{P(\text{both bulls and at least 1 bull})}{P(\text{at least 1 bull})} = \frac{1/4}{3/4} = \frac{1}{3}$$

♢

Example 2.21 (Two Offspring part 2)**.** Suppose that Cindy the Cow has two calves and we know that the first born is a bull. What is the probability that both calves are bulls?

$$P(\text{both bulls} \mid \text{first is a bull}) = \frac{P(\text{both bulls and first is a bull})}{P(\text{first is a bull})} = \frac{1/4}{2/4} = \boxed{1/2}$$

♢

Lemma 2.4 (Multiplicative rule)**.** *Let A, B be two events. If $\mathbb{P}(B) > 0$,*

$$\mathbb{P}(A \cap B) = \mathbb{P}(A|B)\mathbb{P}(B).$$

Further, let A_i be a sequence of events such that $\mathbb{P}(\cap_{i=1}^{n-1} A_i) > 0$. Then,

$$\mathbb{P}(\cap_{i=1}^{n} A_i) = \mathbb{P}(A_1)\mathbb{P}(A_2|A_1)\mathbb{P}(A_3|A_1 \cap A_2)\ldots\mathbb{P}(A_n|\cap_{i=1}^{n-1} A_i).$$

Proof. Proof by induction. First verify the case $n = 2$ holds. Then suppose the case $n - 1$ holds, and show the case n also holds.

$[n=2]$ $P(A|B) = \frac{P(A \cap B)}{P(B)} \Rightarrow P(A \cap B) = P(A|B)P(B)$ ✓

Now assume $P(A_1 \cap A_2 \cap \cdots \cap A_{n-1}) = P(A_1)P(A_2|A_1)P(A_3|A_2 \cap A_1) \cdots P(A_{n-1} \mid \bigcap_{i=1}^{n-2} A_i)$ ☆

Then $P(A_1 \cap \cdots \cap A_n) = P(A_n \mid \bigcap_{i=1}^{n-1} A_i) \cdot P(\bigcap_{i=1}^{n-1} A_i)$ ☆

$= P(A_1)P(A_2|A_1)P(A_3|A_2 \cap A_1) \cdots P(A_{n-1} \mid \bigcap_{i=1}^{n-2} A_i) \, P(A_n \mid \bigcap_{i=1}^{n-1} A_i)$ ☆

□

Theorem 2.5 (Bayes rule)**.** *Let A, B be two events. If $\mathbb{P}(B) > 0$,*

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)},$$

where the denominator $\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A) + \mathbb{P}(B|A^c)\mathbb{P}(A^c)$.

Proof. By definition of conditional probability and the multiplicative rule.

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

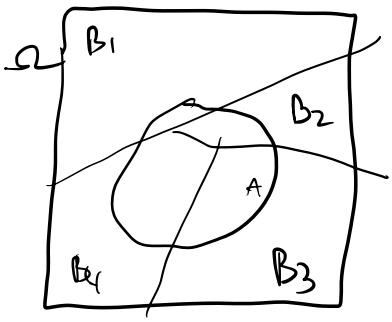
□

We will revisit this more in detail in the next section.

Theorem 2.6 (The law of total probability)**.** *Suppose events $\{B_j\}$ form a partition of the sample space Ω and $\mathbb{P}(B_j) > 0$ for all $j = 1, ..., k$. Then, for every event $A \subset \Omega$,*

$$\mathbb{P}(A) = \sum_{j=1}^{k} \mathbb{P}(B_j)\mathbb{P}(A|B_j).$$

Proof. By the definition of partition and the multiplicative rule.



$A = \bigcup_{j=1}^{k} (A \cap B_j)$ and $(A \cap B_j)$ are disjoint

Then $P(A) = \sum_{j=1}^{k} P(A \cap B_j) = \sum_{j=1}^{k} P(A|B_j)P(B_j)$

□

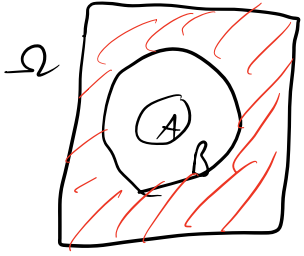
The number of partitions k is allowed to be infinite.

There is a "conditional" version of the law of total probability:

$$\mathbb{P}(A|C) = \sum_{j=1}^{k} \mathbb{P}(B_j|C)\mathbb{P}(A|B_j \cap C).$$







If $A \subset B$ with $\mathbb{P}(B) > 0$, $\mathbb{P}(A|B) =$ $\frac{P(A)}{P(B)}$.
If A and B are disjoint events with $\mathbb{P}(B) > 0$, $\mathbb{P}(A|B) =$ 0.
If $\mathbb{P}(A) > 0$, $\mathbb{P}(A|A) =$ 1

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

If $A \subset B$ then $P(A \cap B) = P(A)$

If A and B are disjoint then $P(A \cap B) = 0$

$$P(A|A) = \frac{P(A \cap A)}{P(A)} = \frac{P(A)}{P(A)} = 1$$

Example 2.22 (Rare disease)**.** Overall disease rate: $\mathbb{P}(D = 1) = 1\%$. The test's accuracy is 95%: $\mathbb{P}(T = 1|D = 1) = \mathbb{P}(T = 0|D = 0) = 0.95$.

- Suppose John Doe tests positive, what is the probability that he actually has the disease?
- Suppose John Doe tests negative, what is the probability that he does not have the disease?
- Suppose John Doe is tests negative, what is the probability that he have the disease?

1) $$P(D=1 \mid T=1) = \frac{P(T=1 \mid D=1)P(D=1)}{P(T=1)} = \frac{0.95 \times 0.01}{P(T=1)}$$

but $$P(T=1) = P(T=1 \mid D=1)P(D=1) + P(T=1 \mid D=0)P(D=0)$$

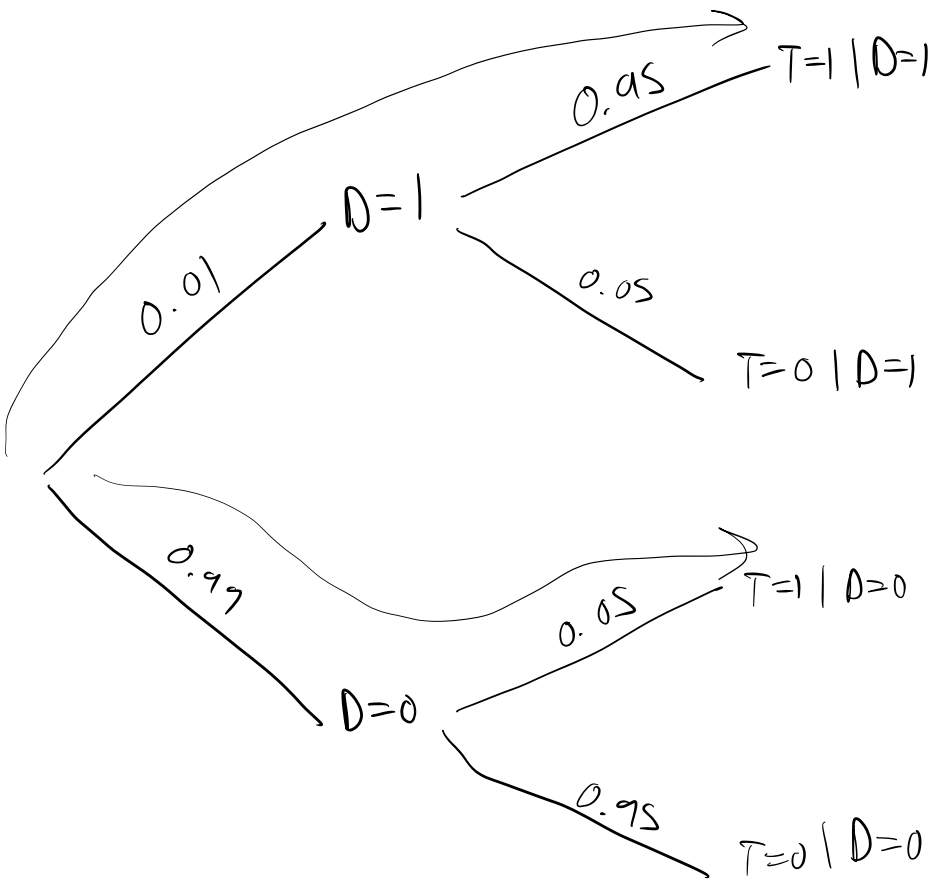
$$= 0.95 \times 0.01 + 0.05 \times 0.99$$

so $$P(D=1 \mid T=1) = \frac{0.95 \times 0.01}{0.95 \times 0.01 + 0.05 \times 0.99}$$

2) $$P(D=0 \mid T=0) = \frac{P(T=0 \mid D=0)P(D=0)}{P(T=0)} = \frac{0.95 \times 0.99}{P(T=0)}$$

and $$P(T=0) = 1 - P(T=1)$$

3) $$P(D=1 \mid T=0) = 1 - P(D=0 \mid T=0)$$


0.01
D=1
0.95
T=1 | D=1
0.05
T=0 | D=1
0.99
D=0
0.05
T=1 | D=0
0.95
T=0 | D=0